Student Performance Prediction Pipeline

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An end-to-end ML pipeline classifying 7,000 students into Low, Medium, and High performance tiers — built for scale and investor insight.

EDA & Preprocessing

Data foundation: 7,000 records, clean pipeline

7,000

Structured socio-academic dataset covering parental education, lunch type, test preparation, and prior scores

3-step

Median imputation for missing values → IQR outlier removal → Label Encoding for categorical features

100%

Clean, normalized, and encoded dataset — no data leakage, no distribution distortion

01 Score averaging

Math, Reading, and Writing scores averaged into a single composite performance score.

02 Class labeling

Composite score thresholds define three performance tiers: Low, Medium, and High.

03 Train/test split

Dataset split before any oversampling to prevent data leakage into evaluation.

04 SMOTE applied

Synthetic Minority Oversampling applied to training data only, balancing all three classes for unbiased learning.

Random Forest Classifier
Built for robustness and interpretability

200 Trees
Ensemble of 200 decision trees reduces single-tree overfitting and improves generalization across all three performance classes

Max Depth 10
Depth cap limits over-complex branches, controlling variance while retaining predictive power

StandardScaler
All input features normalized to zero-mean, unit-variance before training - ensures no feature dominates due to scale

Results: 68.7% test accuracy — optimization roadmap to 80%+

Test accuracy — validated on held-out test split across all three performance classes

68.7%

Cross-val score — 5-fold validation confirms consistent generalization, not a one-time result

70%

Train accuracy — strong model fit shows the pipeline is learning meaningful patterns

89%

Overfitting gap — train−test delta signals a regularization opportunity: tuning min_samples_leaf and max_features is the next step

~20pt